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### PCCT APPARATUS AND CONTROL METHOD THEREOF

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#### Abstract

Provided are a PCCT apparatus and a control method thereof capable of improving correction accuracy of a projection value even in a range in which the projection value is smaller. A PCCT apparatus includes a scanner that rotates an X-ray source which irradiates a subject with X-rays and a photon counting detector which detects X-rays transmitted through the subject for each of a plurality of energy bins, around the subject, an image generation unit that generates a tomographic image by using projection data calculated based on an output of the photon counting detector, and a controller that controls each unit, in which the controller calculates a conversion expression for converting a projection value of the subject into a transmission length by using a plurality of projection values acquired by performing rotational imaging on a phantom which is disposed not to overlap a rotation center of the scanner and of which a material, a shape, and a size are known, along with a plurality of transmission lengths of the phantom.

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## **Background/Summary**

### **CLAIM OF PRIORITY**

[0001] The present application claims priority from Japanese Patent Application JP 2024-033095 filed on Mar. 5, 2024, the content of which is hereby incorporated by reference into this application.

### **BACKGROUND OF THE INVENTION**

#### **1. Field of the Invention**

[0002] The present invention relates to a photon counting computed tomography (PCCT) apparatus that generates a tomographic image of a subject by using projection data divided into a plurality of energy bins, and particularly relates to a technique of correcting non-linearity of a projection value.

#### **2. Description of the Related Art**

[0003] An X-ray CT apparatus uses projection data from multiple directions that is obtained by rotating, around a subject, an X-ray source that irradiates a subject with X-rays and an X-ray detector that detects the X-rays transmitted through the subject to generate a tomographic image of the subject. The generated tomographic image depicts an organ shape of the subject and is used for image diagnosis.

[0004] Since an X-ray absorption coefficient of substances included in the subject is generally smaller as an X-ray energy is higher, in a case where the emitted X-rays have a predetermined energy width, so-called beam hardening, in which the average energy of the X-rays is increased by the transmission of the X-rays through the subject, occurs. Since the beam hardening deteriorates the image quality of the tomographic image by making a relationship between a projection value obtained from an X-ray dose transmitted through the subject and a transmission length of the subject non-linear, correction for the projection value is required.

[0005] JP2004-180808A discloses that a projection value is corrected by using a correction function obtained by performing beam hardening correction and a plurality of function fittings on a sinogram obtained by performing rotational imaging on a phantom disposed to be shifted from a center of an imaging region.

### **SUMMARY OF THE INVENTION**

[0006] However, in JP2004-180808A, consideration for correction accuracy in a range in which the projection value is smaller, that is, in a range in which the transmitted X-ray dose is larger is insufficient. In a PCCT apparatus including a photon counting detector, which counts incident X-ray photons one by one, as an X-ray detector, a proportional relationship between input and output is broken due to an influence of a pile-up or the like that occurs in a case where a larger number of X-ray photons are detected, that is, in a case where the X-ray dose is larger, and thus the correction accuracy of the projection value is decreased. The decrease in the correction accuracy of the projection value deteriorates the image quality of the tomographic image.

[0007] Therefore, an object of the present invention is to provide a PCCT apparatus and a control method thereof capable of improving correction accuracy of a projection value even in a range in which the projection value is smaller.

[0008] In order to achieve the above object, a PCCT apparatus according to an aspect of the present invention comprises: a scanner that rotates an X-ray source which irradiates a subject with X-rays and a photon counting detector which detects X-rays transmitted through the subject for each of a plurality of energy bins, around the subject; an image generation unit that generates a tomographic image by using projection data calculated based on an output of the photon counting detector; and a controller that controls each unit, in which the controller calculates a conversion expression for converting a projection value of the subject into a transmission length by using a plurality of projection values acquired by performing rotational imaging on a phantom which is disposed not to overlap a rotation center of the scanner and of which a material, a shape, and a size are known, along with a plurality of transmission lengths of the phantom.

[0009] In addition, in the present invention, there is provided a control method of a PCCT apparatus including a scanner that rotates an X-ray source which irradiates a subject with X-rays and a photon counting detector which detects X-rays transmitted through the subject for each of a plurality of energy bins, around the subject, and an image generation unit that generates a tomographic image by using projection data calculated based on an output of the photon counting detector, the control method comprising: calculating a conversion expression for converting a projection value of the subject into a transmission length by using a plurality of projection values acquired by performing rotational imaging on a phantom which is disposed not to overlap a rotation center of the scanner and of which a material, a shape, and a size are known, along with a plurality of transmission lengths of the phantom.

[0010] According to the present invention, it is possible to provide a PCCT apparatus and a control method thereof capable of improving correction accuracy of a projection value even in a range in which the projection value is smaller.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a diagram showing an overall configuration of a PCCT apparatus.

[0012] FIG. 2 is a graph showing an example of X-rays divided into a plurality of energy bins.

[0013] FIG. 3 is a flowchart showing an example of a processing flow in Example 1.

[0014] FIG. 4 is a diagram showing an example of a projection value of a phantom that is disposed not to overlap a rotation center of a scanner.

[0015] FIG. 5 is a diagram illustrating a jig that supports a phantom.

[0016] FIGS. 6A and 6B are diagrams illustrating calculation of a position of a phantom based on projection data.

[0017] FIG. 7 is a diagram illustrating calculation of a position of a phantom based on a tomographic image generated by using projection data.

[0018] FIG. 8 is a diagram illustrating calculation of transmission lengths of a phantom.

[0019] FIG. 9 is a graph illustrating calculation of an expression for converting a projection value into a transmission length.

[0020] FIGS. 10A to 10D are diagrams illustrating a case where the phantom is disposed at a rotation center of the scanner.

[0021] FIGS. 11A to 11D are diagrams illustrating a case where a phantom is disposed on a top plate.

[0022] FIGS. 12A to 12D are diagrams illustrating a case where a phantom having a polygonal cross-sectional shape is used.

### DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0023] Hereinafter, examples of a photon counting computed tomography (PCCT) apparatus according to an embodiment of the present invention will be described with reference to the

accompanying drawings. In the following description and the accompanying drawings, components having the same functional configuration will be denoted by the same reference numerals, and duplicated descriptions will not be repeated.

#### Example 1

[0024] An overall configuration of a PCCT apparatus in Example 1 will be described with reference to FIG. 1. The PCCT apparatus includes a scanner **100** and an operation unit **120**. The scanner **100** is installed in an imaging room surrounded by a shielding material that blocks X-rays, and the operation unit **120** is installed in an operation room located outside the imaging room. A rotation axis direction of the scanner **100** is referred to as a Z-axis, a horizontal direction perpendicular to the Z-axis is referred to as an X-axis, and a vertical direction perpendicular to the Z-axis is referred to as a Y-axis.

[0025] The scanner **100** includes an X-ray source **101**, a rotating plate **102**, a collimator **103**, a photon counting detector **106**, a data collection unit **107**, an examination table **105**, a rotating plate controller **108**, an examination table controller **109**, an X-ray controller **110**, and a high-voltage generation unit **111**. The X-ray source **101** is a device that irradiates a subject **10** with X-rays placed on the examination table **105** and is, for example, an X-ray tube device. The collimator **103** is a device that limits an irradiation range of X-rays. The rotating plate **102** includes an opening portion **104** through which the subject **10** placed on the examination table **105** enters, is equipped with the X-ray source **101** and the photon counting detector **106**, and rotates the X-ray source **101** and the photon counting detector **106**, around the subject **10**.

[0026] The photon counting detector **106** is a device that is disposed to face the X-ray source **101** and acquires a spatial distribution of X-rays by including a plurality of detection elements that detect the X-rays. The detection elements of the photon counting detector **106** are arranged two-dimensionally in a rotation direction and a rotation axis direction of the rotating plate **102**, detect incident X-ray photons one by one, and discriminate each X-ray photon into the plurality of energy bins. FIG. 2 shows the detected X-rays divided into three energy bins which are respectively from T1 to T2, from T2 to T3, and from T3 as bin1, bin2, and bin3. The data collection unit **107** is a device that collects the spatial distribution of the X-rays for each energy bin acquired by the photon counting detector **106** as digital data.

[0027] The rotating plate controller **108** is a device that controls rotation and inclination of the rotating plate **102**. The examination table controller **109** is a device that controls upward, downward, forward, backward, leftward, and rightward movements of the examination table **105**. The high-voltage generation unit **111** is a power source that generates a tube voltage, which is a voltage applied to the X-ray source **101**, and a tube current, which is a current supplied to the X-ray source **101**. The X-ray controller **110** is a device that controls an output of the high-voltage generation unit **111**. The rotating plate controller **108**, the examination table controller **109**, and the X-ray controller **110** are, for example, a micro-processing unit (MPU) or the like.

[0028] The operation unit **120** includes an input unit **121**, an image generation unit **122**, a display unit **125**, a storage unit **123**, and a system controller **124**. The input unit **121** is a device that is used to input examination data such as a name of the subject **10**, an examination date and time, and an imaging condition, and is, for example, a keyboard, a pointing device, a touch panel, or the like. The image generation unit **122** is a device that generates a tomographic image by using digital data collected by the data collection unit **107**, and is, for example, an MPU, a graphics processing unit (GPU), or the like. The display unit **125** is a device that displays the tomographic image or the like generated by the image generation unit **122**, and is, for example, a liquid crystal display, a touch panel, or the like. The storage unit **123** is a device that stores the digital data collected by the data collection unit **107**, the tomographic image generated by the image generation unit **122**, a program to be executed by the system controller **124**, data to be used by the program, and the like, and is, for example, a hard disk drive (HDD), a solid state drive (SSD), or the like. The system controller **124** is a device that controls each unit such as the rotating plate controller **108**, the examination table

controller **109**, and the X-ray controller **110**, and is, for example, a central processing unit (CPU). [0029] With the generation of the tube voltage and the tube current by the high-voltage generation unit **111** based on the imaging condition set via the input unit **121**, the X-rays according to the imaging condition are emitted from the X-ray source **101** to the subject **10**. The photon counting detector **106** detects the X-rays emitted from the X-ray source **101** and transmitted through the subject **10** with multiple detection elements to acquire the spatial distribution of the transmitted X-rays. The rotating plate **102** is controlled by the rotating plate controller **108** and rotates based on the imaging condition input through the input unit **121**, particularly a rotation speed or the like. The examination table **105** is controlled by the examination table controller **109** and moves relative to the rotating plate **102** to move an imaging position designated with respect to the subject **10** to an imaging field of view, which is a range in which the transmitted X-rays are detected.

[0030] With repetition of the emission of the X-rays by the X-ray source **101** and the detection of the X-rays by the photon counting detector **106** together with the rotation of the rotating plate **102**, projection data of the subject **10** is measured at various projection angles. In the projection data, a view representing each projection angle is associated with a channel (ch) number and a column number which are detection element numbers of the photon counting detector **106**. The measured projection data is transmitted to the image generation unit **122**. The image generation unit **122** performs back-projection processing on a plurality of projection data to generate the tomographic image. The generated tomographic image is displayed on the display unit **125** or stored in the storage unit **123** as a medical image.

[0031] Since the X-ray absorption coefficient of substances included in the subject **10** is generally smaller as the X-ray energy is higher, in a case where the X-rays emitted from the X-ray source **101** have a predetermined energy width, so-called beam hardening occurs in the X-rays transmitted through the subject **10**. Since the beam hardening makes a relationship between the projection value obtained from the X-ray dose transmitted through the subject **10** and the transmission length of the subject **10** non-linear, correction for the projection value of the subject **10** is required. In particular, in the PCCT apparatus including the photon counting detector **106**, the correction accuracy in a range in which the projection value is smaller is important because the projection value is further non-linear due to the influence of the pile-up or the like, which occurs in a case where the X-ray dose is larger, in addition to the beam hardening. Therefore, in Example 1, even in a range in which the projection value is smaller, a conversion expression that can convert the projection value of the subject **10** into the transmission length with high accuracy is calculated.

[0032] An example of a processing flow in Example I will be described step by step with reference to FIG. 3.

### S301

[0033] The system controller **124** operates the scanner **100** in a state where nothing is placed in the opening portion **104** to image air. The imaging data of the air is used, for example, to calculate the sensitivity of each detection element of the photon counting detector **106**.

### S302

[0034] As shown in (a) of FIG. 4, the phantom **400** is set in the opening portion **104** not to overlap the rotation center **401** of the scanner **100**. The phantom **400** has a known material, shape, and size. (a) of FIG. 4 illustrates the phantom **400** in which a cross-sectional shape of the phantom **400** is a circle and a diameter of the circle is less than a radius of the opening portion **104**.

[0035] The phantom **400** is supported from the rotation axis direction of the scanner **100** by, for example, a jig **500** shown in FIG. 5. The jig **500** illustrated in FIG. 5 is attached to a distal end of the top plate **105A** on the examination table **105**. The phantom **400** is set in the opening portion **104** by sliding the top plate **105A** in the rotation axis direction of the scanner **100**. In addition, a position of the phantom **400** is adjusted by upward, downward, leftward, and rightward movements of the base **105B** that supports the top plate **105A** from below.

### S303

[0036] The system controller **124** operates the scanner **100** in a state in which the phantom **400** is set in the opening portion **104** to perform rotational imaging on the phantom **400**.

### **S304**

[0037] The system controller **124** calculates projection data of the phantom **400** by using imaging data of the air obtained in **S301** and imaging data of the phantom **400** obtained in **S303**. Prior to calculating the projection data, pre-processing such as offset correction, logarithmic conversion processing, and reference correction may be performed on each piece of imaging data.

[0038] (b) of FIG. **4** illustrates projection data **402** of the phantom **400** set not to overlap the rotation center **401**. The projection data **402** indicates a projection value for each coordinate represented by a projection angle  $\theta$  and a channel number  $ch$  of the detection element. In addition, (c) of FIG. **4** illustrates a projection profile **403** representing a projection value for each projection angle  $\theta$  in a center  $ch$ . In the projection data **402** of the phantom **400** set not to overlap the rotation center **401**, the projection profile **403** of most of the  $ch$  includes a projection value from zero to the maximum value.

### **S305**

[0039] The system controller **124** calculates the position of the phantom **400**. For example, the position of the phantom **400** is calculated based on the amount of upward, downward, leftward, and rightward movement of the base **105B** of the examination table **105**. In addition, the position of the phantom **400** may be calculated based on the projection data **402** calculated in **S304**.

[0040] An example of the calculation of the position of the phantom **400** based on the projection data **402** will be described with reference to FIGS. **6A** and **6B**. First, at each projection angle  $\theta$  of the projection data **402**, a range of  $ch$ , which is a projection value of the air, is extracted. Then, the position of the phantom **400** is calculated by specifying the projection angle at which the range of  $ch$ , which is the projection value of the air, is widest, along with the  $ch$  that is the boundary of the projection value of the air at the projection angle. In FIG. **6A**, the range of  $ch$ , which is the projection value of the air, is indicated by arrows, and it is indicated that the projection angle at which the extracted range of  $ch$  is widest is  $\theta_0$ , and the boundary of the projection value of the air at the projection angle  $\theta_0$  is  $ch_0$ . Since the shape and size of the phantom **400** are known, the position of the phantom **400** is calculated based on the specified  $\theta_0$  and  $ch_0$ .

[0041] In addition, the position of the phantom **400** may be calculated by fitting the boundary of the projection value of the air in the projection data **402** of the phantom **400** with a sine curve. With the calculation of the position of the phantom **400** based on the projection data **402**, the position accuracy is improved as compared with the position of the phantom **400** calculated based on the movement amount of the base **105B**. In addition, the position of the phantom **400** may be calculated based on the tomographic image generated by using the projection data **402**.

[0042] An example of the calculation of the position of the phantom **400** based on the tomographic image generated by using the projection data **402** will be described with reference to FIG. **7**. FIG. **7** illustrates an image in which a phantom region **701** is included in an imaging field of view **700** as the tomographic image generated by using the projection data **402**. Since the non-linearity of the projection data **402** is not corrected, the phantom region **701** includes the ring artifact **702**.

However, since the boundary of the phantom region **701** can be specified, the position of the phantom **400** can be calculated. In addition, with the calculation of the position of the phantom **400** based on the tomographic image, the position accuracy is improved as compared with the position of the phantom **400** calculated based on the movement amount of the base **105B**. Return to the description of FIG. **3**.

### **S306**

[0043] The system controller **124** calculates the transmission length of the phantom **400** at the position calculated in **S305** for each projection angle  $\theta$  and each channel number  $ch$  of the detection element.

[0044] The calculation of the transmission length of the phantom **400** will be described with

reference to FIG. 8. Since the position of the phantom **400** having a known shape and size is required, a length of a portion where a straight line connecting the X-ray source **101** and each detection element of the photon counting detector **106** overlaps the phantom **400** is calculated as the transmission length of the phantom **400**. FIG. 8 shows a transmission length **801** in the detection element of ch1 and a transmission length **802** in the detection element of ch2 in a case where the projection angle is  $\theta 1$ . The transmission length in the detection element of ch3 is zero. S307

[0045] The system controller **124** calculates a conversion expression for converting the projection value of the subject **10** into the transmission length by using the projection data **402** calculated in S304 and the transmission length calculated in S305.

[0046] The calculation of the conversion expression will be described with reference to FIG. 9. The transmission length calculated in S305 and the projection value obtained from the projection data **402** are obtained for each projection angle  $\theta$  and each channel number ch of the detection element, and thus the obtained transmission length and projection value are associated with each other by the projection angle  $\theta$  and the channel number ch. Therefore, as illustrated in FIG. 9, a set of the transmission length and the projection value associated with each other by the projection angle  $\theta$  and the channel number ch is plotted as one measurement point **900** on a coordinate plane in which the projection value is on a horizontal axis and the transmission length is on a vertical axis. FIG. 9 is a graph in which each of measurement points **900** for each of a plurality of projection angles  $\theta$  are plotted at a predetermined channel number ch. That is, each of the measurement points **900** shown in FIG. 9 is a set of a transmission length and a projection value obtained for each of different projection angles  $\theta$  at the same channel number ch.

[0047] Then, a conversion expression for converting the projection value into the transmission length at a predetermined channel number ch is calculated by fitting a plurality of the measurement points **900** illustrated in FIG. 9 with an approximate curve **901** represented by a high-order function or the like. The calculated conversion expression is used in a case where the projection value obtained from the projection data **402** of the subject **10** is converted into the transmission length. The conversion expression may be calculated for each channel number ch, or a common conversion expression may be calculated for all the channel numbers ch.

[0048] Since a conversion expression that can convert the projection value into the transmission length with high accuracy even in a range in which the projection value is smaller is calculated by the processing flow illustrated in FIG. 3, it is possible to suppress image quality deterioration of the tomographic image generated by the PCCT apparatus.

[0049] A comparative example with Example I will be described with reference to FIGS. 10A to 10D. In a case where a center of the phantom **400** overlaps the rotation center **401** of the scanner **100** as shown in FIG. 10A, the projection data **402** of the phantom **400** is as shown in FIG. 10B, and the projection profile **403** indicates a constant projection value regardless of the projection angle  $\theta$  as shown in FIG. 10C. As a result, the measurement point **900** plotted on a coordinate plane in which the projection value is on a horizontal axis and the transmission length is on a vertical axis is only one point as shown in FIG. 10D, and thus the fitting cannot be performed with the approximate curve **901**, and the conversion expression for converting the projection value of the subject **10** into the transmission length cannot be calculated.

[0050] The disposition of the phantom **400** is not limited by the jig **500** illustrated in FIG. 5. A case where the phantom **400** is placed on the top plate **105A** of the examination table **105** will be described with reference to FIGS. 11A to 11D. In a case where the phantom **400** is placed on the top plate **105A** as shown in FIG. 11A, the projection data **402** of the phantom **400** includes a projection value of the top plate **105A**. Therefore, the projection data **402** of the top plate **105A** illustrated in FIG. 11C may be acquired by performing rotational imaging in a state in which only the top plate **105A** is placed in the opening portion **104** as shown in FIG. 11B, and the acquired projection data **402** may be used to calculate the conversion expression.

[0051] For example, a projection profile **1101** of the top plate **105A** obtained from the projection data **402** in FIG. **11C** overlaps the projection profile **403** of the phantom **400** placed on the top plate **105A** to specify a region **1102** including the top plate **105A**. Then, the projection value of the specified region **1102** is not used to calculate the conversion expression. In addition, the projection profile **1101** of the top plate **105A** may be subtracted from the projection profile **403** of the phantom **400** placed on the top plate **105A** to calculate the projection value of only the phantom **400**. By using the projection data **402** of the top plate **105A** to calculate the conversion expression, it is possible to suppress deterioration in correction accuracy of the conversion expression. In addition, by placing the phantom **400** on the top plate **105A**, the jig **500** as shown in FIG. **5** can be made unnecessary.

[0052] The cross-sectional shape of the phantom **400** is not limited to a circle. A case where the phantom **400** having a polygonal cross-sectional shape is used will be described with reference to FIGS. **12A** to **12D**. The projection data **402** illustrated in FIG. **12B** is acquired by rotational imaging in a state in which the phantom **400** having a right-angled triangular cross-sectional shape as shown in FIG. **12A** is disposed not to overlap the rotation center **401**. The projection profile **403** obtained from the projection data **402** in FIG. **12B** includes a region in which the inclination of the projection value with respect to the projection angle  $\theta$  is smaller than that in a case of a circular shape illustrated in (c) of FIG. **4**, as illustrated in FIG. **12C**. As a result, a sampling interval of the projection value is shorter in a range in which the projection value is close to zero, and the measurement points **900** are also densely plotted in a range in which the projection value is close to zero as shown in FIG. **12D**. Therefore, the correction accuracy of the projection value in a range in which the projection value is smaller can be further improved. It should be noted that, as the internal angle of the vertex of the polygonal shape disposed in the vicinity of the rotation center **401** is smaller, the sampling interval of the projection value is shorter in a range in which the projection value is close to zero. Therefore, it is preferable that the cross-sectional shape of the phantom **400** is a polygonal shape having an acute internal angle.

[0053] The examples of the present invention have been described above. The present invention is not limited to the above examples, and the components can be modified and embodied without departing from the spirit of the invention. In addition, a plurality of the components disclosed in the above examples may be combined as appropriate. Further, some components may be deleted from all the components shown in the above examples.

#### EXPLANATION OF REFERENCES

[0054] **10**: subject [0055] **100**: scanner [0056] **101**: X-ray source [0057] **102**: rotating plate [0058] **103**: collimator [0059] **104**: opening portion [0060] **105**: examination table [0061] **105A**: top plate [0062] **105B**: base [0063] **106**: photon counting detector [0064] **107**: data collection unit [0065] **108**: rotating plate controller [0066] **109**: examination table controller [0067] **110**: X-ray controller [0068] **111**: high-voltage generation unit [0069] **120**: operation unit [0070] **121**: input unit [0071] **122**: image generation unit [0072] **123**: storage unit [0073] **124**: system controller [0074] **125**: display unit [0075] **400**: phantom [0076] **401**: rotation center [0077] **402**: projection data [0078] **403**: projection profile [0079] **500**: jig [0080] **700**: imaging field of view [0081] **701**: phantom region [0082] **702**: ring artifact [0083] **801**: transmission length [0084] **802**: transmission length [0085] **900**: measurement point [0086] **901**: approximate curve [0087] **1101**: projection profile [0088] **1102**: region

#### Claims

1. A PCCT apparatus comprising: a scanner that rotates an X-ray source which irradiates a subject with X-rays and a photon counting detector which detects X-rays transmitted through the subject for each of a plurality of energy bins, around the subject; an image generation unit that generates a tomographic image by using projection data calculated based on an output of the photon counting



detector; and a controller that controls each unit, wherein the controller calculates a conversion expression for converting a projection value of the subject into a transmission length by using a plurality of projection values acquired by performing rotational imaging on a phantom which is disposed not to overlap a rotation center of the scanner and of which a material, a shape, and a size are known, along with a plurality of transmission lengths of the phantom.

2. The PCCT apparatus according to claim 1, wherein a cross-sectional shape of the phantom is a polygonal shape having an acute internal angle.

3. The PCCT apparatus according to claim 2, wherein a vertex of the acute internal angle of the phantom is disposed in a vicinity of the rotation center of the scanner.

4. The PCCT apparatus according to claim 1, wherein, in a case where the phantom is placed on a top plate, a projection value including the top plate is not used to calculate the conversion expression.

5. The PCCT apparatus according to claim 1, wherein the controller calculates a position of the phantom by using projection data of the phantom, prior to obtaining the transmission length of the phantom.

6. The PCCT apparatus according to claim 1, wherein the controller calculates a position of the phantom by using a tomographic image generated from projection data of the phantom, prior to obtaining the transmission length of the phantom.

7. A control method of a PCCT apparatus including a scanner that rotates an X-ray source which irradiates a subject with X-rays and a photon counting detector which detects X-rays transmitted through the subject for each of a plurality of energy bins, around the subject, and an image generation unit that generates a tomographic image by using projection data calculated based on an output of the photon counting detector, the control method comprising: calculating a conversion expression for converting a projection value of the subject into a transmission length by using a plurality of projection values acquired by performing rotational imaging on a phantom which is disposed not to overlap a rotation center of the scanner and of which a material, a shape, and a size are known, along with a plurality of transmission lengths of the phantom.

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